## PROCEDURE

- The Vernier caliper is checked for zero errors and error if found is to be noted.
- The sphere is kept in between the jaws of the Vernier caliper and the main scale reading (MSR )is noted.
- Vernier scale division which coincides with some main scale division ( VSC ) is noted. Multiply this VSC by least count (LC) gives Vernier scale reading ( VSR ).
- Add MSR with VSR. This will be the diameter of the sphere.
- Observations are to be recorded for different positions of the sphere and the average value of the diameter is found. From this value radius of the sphere R is calculated.
- Using the known value of the mass of the sphere M and calculated radius of the sphere R the moment of inertia of the given sphere about its diameter can be calculated using the given formula.

## LEAST COUNT ( LC )

One main scale division (MSD) = ..... cm

Number of Vernier scale divisions = .....

$$\text{Least Count ( LC )} = \frac{\text{1 Main Scale Division (MSD)}}{\text{Total Vernier scale divisions}}$$

= ..... cm

## OBSERVATIONS

Zero error =

Zero correction (Z . C) = Zero error × LC

| Sl.No. | MSR $\times 10^{-2}$m | Vernier coincidence VSC (div) | VSR = (VSC × LC) $\times 10^{-2}$m | TR = (MSR +VSR) $\times 10^{-2}$m | Diameter of the sphere = 2R correct reading = (TR ± Z.C) $\times 10^{-2}$m |
|---|---|---|---|---|---|
| 1 | | | | | |
| 2 | | | | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |
| 6 | | | | | |

Mean diameter 2R = ..... m

Radius of the sphere R = ..... m

R = ..... m



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